

A
Mini Project Report
On
FLOOD ALERTING AND RESCUE SYSTEM

Electronics and Communication Engineering

By

P.Mahanth	21841A0405
P.Murali krishna	22845A0406
B.Ajay Kumar goud	22845A0408
S.Ganesh	22845A0410

Under the esteemed guidance of

Mrs K. Aparna
Associate Professor



AURORA'S TECHNOLOGICAL AND RESEARCH INSTITUTE
(Approved by AICTE and Affiliated to JNTU, Hyderabad)
(Accredited by NAAC with 'A' Grade)
Parvathapur, Uppal, Hyderabad-500 098
(2024-25)



AURORA'S TECHNOLOGICAL AND RESEARCH INSTITUTE
(Approved by AICTE and Affiliated to JNTU, Hyderabad)
(Accredited by NAAC with 'A' Grade)
Parvathapur, Uppal, Hyderabad-500 098
(2024-25)

CERTIFICATE

This is to certify that the project report entitled “**FLOOD ALERTING AND RESCUE SYSTEM** ” has been submitted by P.Mahanth,21841A0405, P.Murali Krishna, 22845A0406, B.AjayKumarGoud,22845A0408,S.Ganesh22845A0410,Communication Engineering to the Jawaharlal Nehru Technological University Hyderabad is a record of bonafide work carried out by them under my guidance and supervision. The result embodied in this project report has not been submitted to any other University or Institute for the award of any degree or diploma.

Date:

Internal Guide

Ms.K.Aparna
Associate Professor

Project coordinator

Mr. S. Rajasheker,
Associate Professor

Head of The Department

Mr. Vinod Chavan

Director

Mr Srikanth Jatla

ACKNOWLEDGEMENT

We are profoundly grateful to express our deep sense of gratitude and respect towards our guide, **Mrs.K.Aparna** Designation, **Department of Electronics and Communication Engineering, AURORA'S TECHNOLOGICAL AND RESEARCH INSTITUTE, PARVATHAPUR**, for Her excellent guidance right from selection of project and Her valuable suggestions throughout the project work. We are thankful to Her for giving opportunity to work in the laboratory at any time. Her constant encouragement and support has been the cause for us to success, in completing this mini project in the college. She has given us a tremendous support both technical and moral front.

We are thankful to **Mr. S. Rajashekar**, Associate Professor, Project Coordinator for his valuable suggestions and support in completion of the project.

We are thankful to **Mr Vinod Chavan**, HoD ECE and Project Review Committee members for their valuable suggestions and support in completion of the project.

We are thankful to **Mr. Srikanth Jatla**, Director, **AURORA'S TECHNOLOGICAL AND RESEARCH INSTITUTE** for the support during and till the completion of the project.

We extend our thanks to College Management for their support and encouragement for the success of our mini project.



AURORA'S TECHNOLOGICAL AND RESEARCH INSTITUTE
(Approved by AICTE and Affiliated to JNTU, Hyderabad)
(Accredited by NAAC with 'A' Grade)
Parvathapur, Uppal, Hyderabad-500 098
(2024-25)

DECLARATION

We hereby declare that the work described in this project, entitled “**FLOOD ALERTING AND RESCUE SYSTEM**” which is being submitted by us in partial fulfillment for the award of Bachelor of Technology in Electronics and Communication Engineering to **AURORA'S TECHNOLOGICAL AND RESEARCH INSTITUTE** is the result of investigation carried by us under the guidance of **Mrs.K.Aparna**, Associate Professor, ECE Department.

The work is original and has not been submitted for any degree of this or any other university.

Place: Hyderabad

Date:

P.Mahanth	21841A0405
P.Murali Krishna	22845A0406
B.Ajay Kumar Goud	22845A0408
S.Ganesh	22845A0410

ABSTRACT

Floods are one of the most devastating natural disasters, causing widespread destruction and loss of life. Timely alerts and rapid response are crucial in mitigating the impact of floods. This project proposes a Flood Alerting and Rescue System, designed to detect rising water levels, send early warnings, and facilitate rescue operations efficiently. The system uses sensors to monitor water levels in real-time and GSM modules to disseminate alerts to local authorities and affected populations. The core of the system includes water level sensors, strategically placed in flood-prone areas, which continuously monitor water levels. When a critical threshold is reached, the system automatically sends warning messages via SMS to local authorities and residents. The proposed system aims to reduce casualties by providing early alerts, supporting timely evacuations, and optimizing rescue efforts through automation and data-driven decision making. By leveraging IoT and communication technologies, this system offers a scalable and cost-effective solution to enhance disaster preparedness and response.

CONTENTS

Chapter No.	Chapter Name	Page No.
1	Introduction	1
2	Literature survey	2
	2.1 Existing system	3
	2.2 Proposed system	4
3	Hardware Description	5
	3.1 Arduino UNO	5
	3.1.1 AT Mega328P	6
	3.1.2 Pin Description	7
	3.2 Regulated power supply	14
	3.2.1 Transformer	15
	3.2.2 Rectifier	15
	3.2.3 Filter	15
	3.2.4 Regulator	16
	3.3 Light Emitting Diode	17
	3.3.1 LED construction	17
	3.3.2 LED working	18
	3.4 Buzzer	19
	3.4.1 Working principle of buzzer	19
	3.4.2 Specifications	22
	3.5 GSM	22
	3.5.1 GSM overview	22
	3.5.2 GSM specifications	22
	3.5.3 GSM modem	24
	3.5.4 Architecture of the GSM network	25
	3.5.5 Introduction to AT Commands	26
	3.6 Water level sensors	27
	3.6.1 working of water level sensor	29
4	Software Description	31
	4.1 Programming Arduino	31
	4.2 Arduino Installation	31

	4.3 Program Structure	38
5	Design and Implementation	40
	5.1 Block diagram overview	40
	5.2 Flow chart	41
6	Advantages and Applications	42
	6.1 Advantages	42
	6.2 Applications	42
7	Results	43
	Conclusion and future scope	45
	References	46
	Appendix	47

LIST OF FIGURES

Fig No.	Name of the Figure	Page No.
2.1	Existing flood alerting system	4
2.2	Proposed system	4
3.1	Arduino-Uno Structure	7
3.2	Pin Description	7
3.3	Regulated power supply	15
3.4	Transformer	15
3.5	Rectifier	16
3.6	Filter	16
3.7	Pin out of 7804	16
3.8	RPS	17
3.9	Symbol of LED	17
3.10	LED	18
3.11	Different Shapes of LED	18
3.12	Working of LED	19
3.13	buzzer terminnals	20
3.14	Water Level Circuit using Buzzer	21
3.15	GSM modem	25
3.16	GSM Architecture	26
3.17	Interfacing GSM modem with ARDUINO UNO	27
3.18	Water level sensor	28
3.19	working of water level sensor	29
4.1	USB Cable	32
4.2	A TO MINI-B CABLE	32
4.3	Opening Arduino	33
4.4	Launching Arduino IDE	34
4.5	Arduino Software Opening New File	35
4.6	Selecting File Type	35
4.7	Selecting Board Type	36
4.8	Selecting Serial Port	37
4.9	Uploading Program To Board	37
4.10	Program Structure	39

7.1	circuit is on	43
7.2	first sensor is activated	43
7.3	second sensor activated	44
7.4	alert message sended	44

LIST OF TABLES

Table No.	Name of the table	Page No.
3.1	commands table	27